AP CALCULUS AB - UNIT 1

Guided Notes 1.15: Connecting Limits at Infinity and Horizontal Asymptotes

Name: ______________ **Class:** ______________ **Date:** ______________

Learning Objectives and Essential Question

Essential question: How can we evaluate limits as $x \to \infty$ and $x \to -\infty$?

- Evaluate limits as $x \to \infty$ and $x \to -\infty$.
- Determine horizontal asymptotes from end limits.
- Use degree comparison, dominant powers, radicals, conjugates, and growth hierarchy.
- Analyze the two ends independently and handle $\sqrt{x^2} = |x|$ correctly.

Key Knowledge, Vocabulary, and Notation

Right end	$x \to \infty$ asks about large positive inputs.
Left end	$x \to -\infty$ asks about large negative inputs.
Horizontal asymptote	If an end limit is L, then $y = L$ is a horizontal asymptote for that end.
Rational degree rule	Lower numerator degree gives 0; equal degrees give leading-coefficient ratio; higher numerator degree is usually unbounded.
Radical warning	$\sqrt{x^2} = |x|$, equal to x on the right and $-x$ on the left.

Concept Development and Strategy

1. For rational functions divide by the highest power in the denominator or compare leading terms.
2. For radical differences, rationalize before comparing dominant terms.
3. A curve may cross a horizontal asymptote; an asymptote describes end behavior, not a barrier.

Worked Example: Equal degrees

Evaluate $\lim_{x\to\pm\infty} \frac{3x^2-x}{2x^2+5}$.

Answer and Reasoning

Dividing by x^2 gives a limit of $3/2$ at both ends, so $y = 3/2$ is a horizontal asymptote.

Worked Example: Different radical ends

Evaluate $\lim_{x\to\pm\infty} \frac{\sqrt{x^2+1}}{x}$.

Answer and Reasoning

Write $\sqrt{x^2+1} = |x|\sqrt{1+1/x^2}$. The right limit is 1 and the left limit is -1.

Worked Example: Conjugate at infinity

Evaluate $\lim_{x\to\infty}(\sqrt{x^2+4x}-x)$.

Answer and Reasoning

Rationalize: $\frac{4x}{\sqrt{x^2+4x}+x} = \frac{4}{\sqrt{1+4/x}+1} \to 2$.

Guided Practice and Discussion

Directions

Show the mathematical setup, preserve exact values unless a decimal is requested, and justify existence or interpretation statements with the appropriate definition or theorem.

1. Evaluate $\lim_{x\to\infty} \frac{3x+1}{x-4}$.

2. Evaluate $\lim_{x\to-\infty} \frac{2x^2-5}{x^2+7}$.

3. Evaluate $\lim_{x\to\pm\infty} \frac{5x^3-x}{2x^3+1}$.

4. Evaluate $\lim_{x\to\pm\infty} \frac{4-3x^2}{x^3+1}$.

5. Evaluate $\lim_{x\to\pm\infty} \frac{2x^2+1}{\sqrt{4x^4+3}}$.

6. Evaluate $\lim_{x\to\infty} x\left(\sqrt{x^2+1}-x\right)$.

7. Construct a function with right horizontal asymptote $y = 2$ and left horizontal asymptote $y = -3$.

AP Reasoning Check

Multiple-Choice Check

Choose the best answer. Explain why at least one distractor is incorrect during class discussion.

1. $\lim_{x\to\infty} \frac{4x^2+1}{2x^2-3x}$ equals

(A) 0 **(B)** 2
(C) 4 **(D)** $+\infty$

2. $\lim_{x\to-\infty} \frac{\sqrt{x^2+1}}{x}$ equals

(A) -1 **(B)** 0
(C) 1 **(D)** $+\infty$

3. A calculator table for $f(x) = \frac{2x^3+1}{x^3-4}$ at large positive and negative inputs approaches

(A) -2 **(B)** 0
(C) 2 **(D)** $+\infty$

Common Errors and AP Precision

- Keep the value of a function at a point separate from its limiting behavior near that point.
- State one-sided behavior explicitly whenever a two-sided limit or continuity claim depends on agreement from both sides.
- A complete AP justification names the relevant definition, condition, or theorem rather than reporting only a numerical result.

Exit Ticket

Prove that a rational function can have at most one finite horizontal asymptote, the same at both ends, when numerator and denominator have equal degree.